Integrating Predictive and Descriptive Analytics to Identify High-Risk Jobs in the Era of Artificial Intelligence

MIP Dissanayaka
Department of languages
Sir John Kotelawala Defence University
Rathmalana , Sri Lanaka
41-adc0005@kdu.ac.lk

WMD Subhashwaree
Department of languages
Sir John Kotelawala Defence University
Rathmalana , Sri Lanaka
41-adc0015@kdu.ac.lk

JANI Jayawardhana
Department of languages
Sir John Kotelawala Defence University
Rathmalana , Sri Lanaka
41-adc0031@kdu.ac.lk

Abstract

Artificial Intelligence (AI) is redefining the workforce of the entire world, since it automates the usual tasks, supplements decision-making methods, and opens new possibilities in new occupational roles. The knowledge of the industries and the types of jobs that AI-based automation affects the most is crucial to the policymakers, organizations, and employees who want to navigate the altering labor market. This work explores the automation threat in different sectors and occupations and determines the new skills and roles that are led by AI technologies. On a curated set of job listings with details about job titles and industries and the sizes and locations of given companies, the level of AI adoption, automation risk, needed skills, and expected wages, we used machine learning techniques to identify automation risk (High vs Not High) and make observations.

An extensive predictive modeling pipeline was applied, which included text features (job titles and required skills), categorical features (industry, company size, location), and numerical features (level of AI adoption, salary, growth forecasts of job). Several machine learning models among them were evaluated, and they were Logistic Regression, Random Forest, Light GBM, XG Boost and a Stacking Ensemble where the evaluation metrics were accuracy and F1-score. The ensemble that was most successful in predictions with 66 percent accuracy was the stacking ensemble, which indicates the strength of using tree-based models together with logistic regression.

It was analyzed that the highest predicted risk of automation is in the Entertainment, Education, and Transportation sectors, but the risk of automation in the healthcare, finance and retail sectors is relatively low. Routine administrative and support jobs were found to be the most vulnerable to job-level predictions. In their turn, new jobs in the AI adoption fields are such as data scientists, machine learning engineers, AI product experts, and cloud computing experts. Python, data analysis, machine learning, natural language processing, and AI tools are becoming more popular among required skills. The findings can be used to make actionable insights in workforce planning, reskilling programs, and organizational strategy concerning the labor market changes due to AI.

Keywords - AI-driven automation, Workforce transformation, Machine learning prediction, Industry-level risk assessment, Emerging job roles, Skills demand forecasting, Job market analytics, Predictive modeling, Natural language processing (for skills text), Stacking ensemble models

1. Introduction

The emergence of Artificial Intelligence (AI) has become one of the most disruptive forces in the restructuring of industries in the world. The current contributions of AI technologies, including machine learning, natural language processing, computer vision, and intelligent automation, are contributing to both savings and the potential of these technologies to replace workers in the labor marketplace. Companies are turning to the use of AI systems to do routine tasks, streamline processes, and improve decision-making. Although automation can decrease the cost of operation and enhancing productivity, it leaves workers with uncertainties because their functional roles become vulnerable to change by AI.

Recent research has pointed out that the level of automation risk is highly dynamic based at industries and types of jobs. As an example, financial, administrative and manufacturing positions are more prone to vulnerability because of repetitive nature of job whereas creative and healthcare related jobs are comparatively less vulnerable. Moreover, AI implementation is a disruptive force, as well as a catalyst of new opportunities, that demand new skills and offers new positions, including machine learning engineers, data analysts, AI consultants, and cloud infrastructure experts. The dynamics are important in that governments, businesses, and educational institutions understand how to design reskilling and workforce development strategies.

Although there is extensive literature on the automation topic, there is a lack of research on the quantitative study of industry-level and job-level AI risk based on real job market data and machine learning models. Majority of the previous research uses macroeconomic forecasts or task surveys but fails to utilize job posting data at the job level. In this study, the authors fill this gap by using a predictive machine-based model on a filtered sample of global job advertisements, using textual (job titles and skills), categorical (industry, company-size, location), and numeric predictors (AI adoption, salary estimates). The main goals of the research are:

1.Determine industries and occupations where AI-based automation can best apply.

2.Estimate the risk of high automation of individual job postings.

3.Evaluate new job descriptions and skills with the implementation of AI.

In Sri Lanka, the adoption of AI in the workforce is progressing slowly and unevenly across different sectors. According to the Department of Census and Statistics of Sri Lanka,[29] the total labor force in 2024 was about 8.5 million. Of these, 47% worked in services, 28% in industry, and 25% in agriculture. The Information and Communication Technology Agency (ICTA) estimates that AI technologies could add almost USD 3.5 billion to the national GDP by 2030. This growth will primarily come from automation and digital changes in finance, healthcare, logistics, and manufacturing.[30] However, a recent analysis by the World Bank warns that around 23% of jobs in Sri Lanka could be at high risk of automation within the next decade.[31] This is especially true for clerical, administrative, and manufacturing positions. On the other hand, new roles like data scientists, AI developers, automation engineers, and digital policy analysts are expected to be in higher demand. Despite these shifts, the Ministry of Technology reports that over 60% of university graduates do not have the digital and analytical skills needed for jobs that involve AI.[32] Therefore, it is important to recognize which industries and jobs are most at risk from AI automation. It is also essential to identify the new skills and job types that will emerge. This understanding will help shape workforce development policies in Sri Lanka and ensure that everyone can take part in an AI-driven economy.

4.Offer practical information to workforce planning and policy interventions.

This work has an added value due to the integration of text analytics, machine learning modeling, and data visualization in providing useful insights to decision-makers. Contrary to other literature, which has mainly evaluated the risk of automation on a macro-level level, the study offers granular, data-driven forecasts on a job by-job basis, allowing the targeting of risked industries and jobs, and skills to focus on in new opportunities.

2. Literature Review

2.1 Introduction

Artificial Intelligence (AI) has been introduced in various industries, which has flipped the world of work. Even though the automation driven by AI can potentially be beneficial (e.g., more efficient and productive), it also causes such a problem as the job displacement and the emergence of new jobs. This literature analyzes the current studies on the impacts of AI-based automation on the industries and jobs to determine the most affected areas and the new jobs and skills that have emerged due to AI-based automation.

2.2 AI and Job Automation: Dangers and Reality.

The literature holds that AI-based automation has an impact on the jobs that have a routine, repetitive or predictable character. Frey and Osborne [1] have noted that there is a threat of automation of as much as 47 per cent of employment in the United States within a 20-year period. Sectors of the production sphere, trade, transportation, and administrative services become the place of great vulnerability due to the characteristics of organizational processes that can easily be replaced by calculation and robotics.

However, some authors like Arntz, Gregory, and Zierahn [2] caution about generalization and they hold that when estimating the automation potential, one may consider jobs as being task-based not as job titles. In their production, they depict that even in high-stake business, not all humanistic and non-hard parts of skills can be substituted, such as judgment, empathy, adaptability, and many others.

The IMF Working Paper (2024) estimates that a maximum of 8 percent of working hours in developed countries can be automated by 2030, with STEM and professional occupations being the most vulnerable to automation, up to 16 percent; because of economic pressures not the entirety of potential automation will be replaced at scale [3]. Similarly, in terms of industrial AI in the manufacturing sector, it was found out that

displacement of jobs and the upskilling demands has been eminent especially in logistics, predictive maintenance or process control tasks [4]. This is due to the emergence of AI algorithms and computing changing many industries, including finance, healthcare, education, and manufacturing. Higher numbers of jobs are created with the help of AI, but the routine and work-intensive jobs are displaced; strain on the labor market is created. This paper describes and explains the danger of automation by AI in the form of job vulnerability by classifying and clustering jobs based on their risk level (high, medium, and low risk jobs). Such outcomes are workforce planning that is practical, reskilling, policy intervention recommendations in an AI-based economy [5].

Recent reports indicate that AI innovation can replace big percentage of the workforce with research indicating that the International Monetary Fund (IMF 2024) report has indicated that AI will touch 40 per cent of all jobs in the global level, and various jobs are at risk of being outsourced. [21]

This opinion does not contradict the predictions that 30 percent of the current jobs will have been fully automated in the U.S by 2030, and 60 percent will have undergone a significant job change in the job market due to the introduction of AI [22].

In advanced economies, the IMF Working Paper (2024) estimates up to 8 percent of working hours can be automated by 2030 and STEM and professional jobs are most at risk of automation and reaching up to 16 percent; due to the strains on the economy, not a single fraction of the potentially automatable jobs will be automated at scale. New systematical reviews are indicating that AI will impact 40 percent of all jobs across the globe, and a significant number of jobs will be in danger of disappearance, which is pointing to the reality on the ground than to the future which will pose a challenge of losing a job [21].

2.3 AI-Based Automation and its Implication on Industries.

The effects of the AI and automation technologies vary across industries. A 2025 study performed by Wang revealed that manufacturing professionals expressed diverse concerns regarding job displacement by AI regardless of industry and job level [6]. Still, on the same topic, Zirar (2023) notes that the application of AI in the workplace has altered 82 per cent of the work positions and the skills required which means that the shift in the various industries is massive [7].

Similarly, as Zirar (2023) notes, the application of AI in the workplace has changed 82 per cent of jobs and the required skills, which means that the changes are enormous in various industries [8].

2.4 Job Displacement

Another question that has been brought up by the introduction of AI and machine learning is the problem of job displacement and how employment can negatively impact it. The article by Tiwari (2023) outlines the current research regarding AI and ML in job displacement and the labor market and has discussions about how the technologies can potentially impact different industries and occupations and whether they would result in the generation of new jobs [9]. The rapid advancement of AI technologies transforms the workplace of the world, affecting the trends of jobs and the requirements of the skills. The paper evaluates the impact of AI against job replacement, skills reorientation, and newly created opportunities by relying on empirical data and literature. It identifies the tendencies in the workforce and provides the professionals with ideas of how to adapt to working with AI and utilize it to innovate and develop [10].

Georgieff (2022) explains that the application of AI can increase labor productivity and output and counter the potential displacement effect of automation on highly digitally skilled workers. This demonstrates that digital literacy is pertinent in mitigating the evil effects of AI on employment [11].2.5 Job Creation and Transformation through AI

2.5 Creation and Transformation of Jobs with the help of AI.

In the same manner that we read in regular tales of job loss, we also read about the AI stimulating the emergence of employment. Brynjolfsson and McAfee [12] assure us that AI does not deprive individuals, on the contrary, it empowers them. And this is where hybrid jobs such as AI ethicists, data analysts, machine learning engineers, and human-centered AI designers come in. The responsibilities require a combination of technical ability and mastery in a profession, a pointer that the digital economy is developing towards multidisciplinary workers.

The concept is further supported by the World Economic Forum Report [13] which projects the automation of 85 million job opportunities by 2025, but again, about 97 million new employment opportunities may arise that further promotes the human and machine division of labor, and algorithms. Such positions will be concentrated in the sphere of healthcare, online communication, education, and AI creation.

2.6 Dexterity Shifts in the Artificial Intelligence Era.

The transformation of the job market is not only through the aspect of the job replacement, or creation, but also through the alteration of the nature of the asking skills required. According to the research of McKinsey Global Institute [14], three key categories of skills shifts exist; the physical to cognitive and social-emotional skills; the routine to analytical work and simple digital literacy to advanced technological skills.

Moreover, the OECD [15] focuses on the problem of constantly growing importance of lifelong learning and continuous upskilling. The employees now lack time to develop their competencies in terms of learning new skills such as techniques of solving problems, digital tools, and collaboration with AI systems among others. These trends denote that the wave of reforms in educational systems and training programs in the field of workforce development must comply with the reality of the new world of AI.

2.7 Next Generation Job Roles and Skills.

New areas of professional orientation and professional competencies are also being opened as AI technologies advance further. Ejjami (2024) addresses the appearance of the new professions that are conditioned by the incautious adoption of artificial intelligence, machine learning, robots, and big data analytics in significant areas, i.e., AI, healthcare, energy, education, and retail. The paper examines the issue of the changing workforce and skills gaps due to the influence of the AI on the operations and decision-making [16]. The reskilling of employees has become a very critical issue to companies because AI is transforming the industries and companies should be able to keep up with the competitive world. The problems in reskilling that the paper will discuss include rapid paced technology, resistance and skills mismatch and provide examples of best practices, which include tailored learning, collaboration with educational facilities,

and continuous learning. Its application of case studies and studies is practical information that can be adopted to implement effective reskilling programs to ensure the workforce is relevant in the age of AI [17]].

Johnson (2025) considers the role of AI in IT as job displacing and offering specialized tasks and new skills to be competitive in the constantly changing world. The discussion states the changes in employee relations in the context of introducing AI into the field of software development, IT support, and cybersecurity protection, based on recent academic research and publications in the industry [18].

2.8 Implications of Policy and Future Directions.

The creation of AI technologies is accelerated, and this is why there is some critical necessity to interfere in policies to resolve the issue of automation. Broady (2025) explores the impacts of the pandemic-related automation and AI burst on the US in 20192022 that contributed to the rising levels of unemployment. The paper emphasizes the need of the policies that can help in the increased mobility of labor and minimization of the harmful impact of AI in the job-market [19].

Subaveerapandiyan (2024) presents a comprehensive bibliometric analysis of the scientific research on the issue of artificial intelligence, automation, and job displacement in 1984-2024. The analysis has demonstrated that there are positive trends of growth of publications, citation rate, and network of researchers and institutions, which can help us to know the evolving scenario in AI-related studies [20].

2.9 Current Job Dislocation trends and Economic Impact.

The growth of the use of AI due to the pandemic has laid down new patterns of labor displacement. Broady et al. (2025) conducted research on the automation and AI intensification process in 2019-2022 and the impact of these events on the labor market in the United States, where they resulted in increased unemployment levels and economic instability in some sectors [24].

Not all effects of the economic impacts are as simple as a job replacement. It is shown that AI will replace the similar amount of 300 million full-time jobs in the world that can include a quarter of the working activity in the US and Europe and, at the same time, will introduce new opportunities and productivity increases. This kind of dual impact brings about the necessity of identifying the pattern of displacement in different spheres of the economy in a sensitive manner [25].

2.10 Workforce Requirements and Skills Transformation.

The tendencies of the labor market that the AI is endangering are not merely those of the displacement of jobs or the creation of new ones but are those of the changes in the very concepts of the skills that are supposed to be possessed by the employees. Recent research points out that about 40 percent of the workforce will require reskilling within a span of six months and half of the workforce requires a complete overhaul on the skillset [26].

The skills of the workers are influenced by the change as they are upskilled and reskilled in the organizations. Some studies indicate that the knowledge of how AI affects the professional competencies, and the implementation issues should be incorporated in the development of the successful skill transformation programs [27]. Companies have been adopting cross-collaborative, scaled workforce development practices due to the realization that the old training paradigms cannot be applied in the times of AI. [28].

2.11 Gaps in the Current Research

Although the existing body of knowledge reveals sufficient information on the theoretical impacts of the AI on the job market, there is a clear dearth of the empirical, data-based studies, which may help define the level of AI impact on the job industry and job occupations. It is projected, simulated, or surveyed instead of occupational data of finer grain, like that of occupational risk to automation, occupational shift, and job trend.

The current paper attempts to address that gap, since it is based on a more comprehensive database, by the degree of AI effect, the threat of job automation, as well as the new skills being added to the pool in an extensive workforce. By doing so, the study will hopefully illuminate the practical information about reorganizing different spheres through the assistance of AI and.

what are the skills that are being particularly useful as regards future. On the one hand, theoretical literature is replete with theoretical and highly generic studies; on the other, there is a gap in literature on sector-specific as well as data-driven studies of the automation risk scores, how they have evolved over time, and the novel skills required which is precisely what your information is equipped to provide. There is low integration of the individual job level degree of automation probability with unveiled advancing in posts and fresh needs of job and skills, and your study might add to it.

2.12 Conclusion

It has been noted in literature that automation with AI can significantly affect different sectors and types of jobs. Some industries have mass displacement of jobs because of automation, but on the other hand, some industries are seeing the creation of new positions which are demanding skilled labor. The results of the research support the need to reskill and upskill the workforce with the changing demands of the work environment. The future studies need to be conducted via longitudinal studies to determine the long-term impacts of AI on employment and what can be done to prevent the negative results.

3. Methodology

3.1 Dataset collection

The article is founded on the research on the so-called AI Job Market Insights Dataset that is gathered on the publicly available source such as Kaggle datasets, which are known to provide high-quality and structured data on data science research. This data set summarizes all job market characteristics associated with the role of artificial intelligence (AI) in the employment market. It offers an understanding of the two-fold effect of AI: the automation threat to the current jobs and the introduction of new work positions and competencies. The data table includes more than 500 records in various industries, such as healthcare industry, financial sector, manufacturing, education, and information technology.

It has ten major variables such as Job Title (a categorical variable, describing the job: e.g. IT, Healthcare), Company Size (categorical, e.g., Small, Medium, Large), AI Adoption Level (categorical, i.e. it is implemented in the job: Low, Medium, High), Automation risk (a percentage) (numerical, the probability that the job may be automated by AI), Skills required (textual, enlisting the key competencies), Job Growth Projection (numerical, future potential of the job being automated by AI

The dataset would include a combination of a text-based (e.g., job titles, skills required), categorical (e.g. level of AI adoption), and numerical variables (e.g. automation risk, salary). The source data was then properly cleaned and preprocessed: median values in the salary and experience columns were used to fill in missing numerical values; outliers in salary were removed with the Interquartile Range (IQR) technique; and categorical variables were coded as integers to be used in predictive modeling. The NLP techniques of TF-IDF and, if present, sentence embeddings by SBERT were used to process textual data like job titles and skills.

This organized dataset allows a systematic exploration of: (i) what industries and job categories are most at risk of automated by AI, found using predictive modeling and automation risk groupings, and (ii) what new jobs and skills are in demand, found by trend analysis and text-based feature extraction. It offers a strong base in the assessment of the transformational effects of AI on the future workforce.

3.2 Data processing

Data processing included a logical sequence of tasks that were necessary to make the data clean, consistent, and machine learning ready. The raw data, aijobmarketinsights.csv, with the 500 observations and 10 features, was initially loaded into the Python environment with the help of the Pandas library. The preliminary exploratory data analysis revealed that no values were absent in all columns, and there was completeness of data. The pipeline thereof automatically identified the column of Required Skills with the aid of a custom detection function, which contributed towards isolating the textual attributes associated with skill requirements. The skill descriptions were refined to increase textual uniformity by changing all characters to lowercase and all punctuations, special characters, and unnecessary white space were deleted.

Automation Risk, the target variable was coded into a binary numerical system to be classified with the help of

modeling. Jobs with a high label got a 1 value and other jobs were approached as 0 as the non-high-risk jobs. This change transformed this problem into a two-category classification problem in terms of high and low automation susceptibility. Also, Job Title and Required Skills were merged into the single text field to bring out more contextual information that would be used to extract features of the text later.

To have a good model evaluation, the dataset was randomly shuffled and divided into those that can be used as training (80) and testing (20) factors with stratified sampling ensuring that the percentage of high- and low-risk jobs in these groups was the same. This gave 400 training and 100 testing records. Normalization of features and categorical encoding of features were also done by the pipeline to unify all the variable types. Categorical variables like Industry, Company Size, and Location were numerically coded over an Ordinal Encoder which enabled the models to make sense of categorical differences without creating artificial magnitude. Numerical variables like AI Adoption Level, Job Growth Projection, and Salaries were normalized with the help of a StandardAero to reduce the scale bias and enhance model convergence. AI Adoption Level feature was further converted into numbers by trying to map qualitative labels, Low, Medium and High to 1, 2 and 3 respectively.

The textual data in job titles and skills were vectorized with the TF-IDF (Term Frequency-Inverse Document Frequency) method in situations where deep language embeddings were not available, and then dimensionality reduction was done with Truncated Singular Value Decomposition (SVD). This minimized both noise and computation and retained critical semantic patterns. The resulting final preprocessed data merged text embeddings, coded categorical data and scaled numerical data in one structured feature matrix, which constitutes a clean and balanced input to the machine learning step.

3.3 Feature Engineering

The feature engineering step was aimed at converting the preprocessed data to a structured information format, which can be used to train machine learning models. Since the data set had both free-form and structured attributes, various encoding and transformation tools were used to aptly describe the semantic and categorical and quantitative forms of all job records.

As in the case of textual information, including Job Title and Required Skills, hybrid representation strategy was chosen. The first step taken was the concatenation of these two columns into a single composite text field which records the occupational context, and the skills associated with it. The model was able to learn the relationship between job descriptions and automation risk in a more holistic way only through this unified textual representation. The pipeline did not include Sentence-BERT (SBERT) embeddings, which are generally employed to query deep semantic representation, in favor of a TF-IDF (Term Frequency-Inverse Document Frequency) approach to numerically encode the text. TF-IDF is the most effective in distinguishing the significance of certain words in a corpus by balancing between term frequency and inverse document frequency making it more important to focus on unique keywords. To handle the dimensionality of TF-IDF vectors and overcome the curse of dimensionality, Truncated Singular Value Decomposition (SVD) was later used to truncate the feature space to 80 latent components. The approach maintained critical semantic variance in this dimensionality reduction, enhancing computation efficiency and extrapolation capacity.

Categorical data like Industry, Company Size and Location were categorized or encoded by Ordinal Encoding which converted categorical data labels into an integer form, and those categories that were not seen were encoded with the unseen coded value parameter set to the same value. This made sure that the models would be able to read the categorical differences without distorting them into numerical values.

In the case of numerical variables, they were subjected to special transformations to make the features easier to interpret and the model more accurate. The AI Adoption Level characteristic, which was initially in the form of text (Low, Medium, High), was transformed into numerical ones (1, 2, and 3), which represented a progressive degree of AI implementation in organizations. Moreover, continuous data like Job Growth Projection and Salary USD were quantified and standardized with the help of StandardAero that brought

the values to a similar range so that a large numeric range of one feature does not dictate the model training.

All three domains of engineering, namely, textual, categorical and numerical, were then horizontally concatenated into integrated training and testing matrices. This multidimensional feature space enabled the models to be trained using a variety of features at the same time. The ultimate feature space was successful in incorporating job semantics, industry peculiarities, and quantitative economic data, which can be viewed as a holistic basis of predictive modeling of automation risk in diverse job fields.

3.4 Model Development

The stage of model development meant building and testing several supervised machine learning algorithms to estimate the risk of AI automation of the different jobs. The main task was to create a strong predictive system that would be able to differentiate between high- and low-risk jobs based on a set of textual, categorical, and numerical characteristics based on the preliminary phases.

The dataset was split into the training (80 per cent) and testing (20 per cent) groups to make sure that the models are reliable, and the distribution of the automation risk levels was retained. The Scikit-learn and Light GBM models were used to train and evaluate the model, and the standardized hyperparameters were optimized based on the cross-validation. The pipeline utilized five significant models, namely, Logistic Regression, Random Forest, XG Boost, Light GBM and an Ensemble Stacking Model, each of which brings complementary features in the general prediction process.

Logistic Regression model was used as the baseline classifier because it is readily interpretable, and it makes available probabilistic information regarding the importance of features. It was used as a linear reference point; values of the standardized and coded input features were mapped to the binary risk automation output. It was a basic model, but it formed a valid point of reference regarding models.

Random Forest Classifier was proposed to address nonlinear associations and feature interactions, which the logistic regression was unable to address. Random Forest enabled better prediction stability and overfitting Reduction because of predictions made by multiple decision trees. The important hyperparameters, such as the number of estimators (as 200), and maximum depth (as auto) were adjusted to get the optimal performance without affecting the interpretability.

More accuracy was also achieved by using gradient-boosting algorithms, namely XG Boost and Light GBM, which are efficient in the processing of structured and high-dimensional data. The XG Boost model adopted a tree-based boosting algorithm that minimized the classification error on a series of iterations, whereas Light GBM adopted a faster-computing and better large feature space operation by using histogram-based learning and leaf-wise growth of the tree. The performance of both models was better in replicating complex interactions of features particularly between job skills, level of AI adoption, and type of industry.

Lastly, a Stacking Ensemble Model was created to synthesize the predictive abilities of all the preceding algorithms. Within this framework of meta-model, the final estimator was Logistic Regression which combined the predictions of the base learners (Random Forest, XG Boost and Light GBM). The stacking method also contributed greatly to the generalization performance by exploiting the variation of base models.

A regular evaluation protocol was used through model development with measures like Accuracy, Precision, Recall, and F1-Score used to measure predictive quality. The last ensemble model had the best overall score in these metrics which show that model stacking is effective in both linear and nonlinear association modelling. Also, a ROC-AUC (Receiver Operating Characteristic - Area Under Curve) analysis was made to validate that the ensemble model had better discriminative power by essentially comparing high and low risk job descriptions.

The entire model formulation procedure has not only developed a robust predictive framework but also gave explanatory results into the correlation among adoption of AI, job skill demands, and risk of automation. All these models help in creating a data-driven insight regarding the

transformation of the job market landscape due to technological transformation.

3.5 Analysis Approach

The evaluation and analysis were performed to perform the critical evaluation of the predictive performance, interpretability, and reliability of the developed models. This phase included quantitative analysis based on conventional machine learning metrics and qualitative analysis based on feature importance visualization and analytical information. This was aimed at identifying the extent to which the models could identify occupations that were most likely to be automated and finding out the most critical factors affecting a job forecast.

The evaluation of model performance was conducted on the unseen test set (20%), to make sure that the performance was based on the ability to generalize in real-life. To present a detailed overview of the performance, the evaluation used four main classification measures, such as Accuracy, Precision, Recall, and F1-Score. Accuracy was used in assessing the overall signal of correct prediction whereas precision and recall were used to determine how well the model was able to detect high-risk jobs with no false positive or negative results. The main measure of model comparison was the harmonic mean of precision and recall (F1-Score) because it is resistant to the use of imbalanced data sets.

The individual models with moderate accuracy and limited inability to model complex nonlinear interaction were found with the Logistic Regression. Random Forest and Boost performed better than the baseline model since they deliver balanced results with all metrics because of the ensemble character and the ability to model hierarchical relationships between features. Light GBM model among the single model models showed the best performance, as it enjoyed the advantages of leaf-wise tree construction and high-dimensional data in the form of TF-IDF features can be effectively dealt with. Nevertheless, the Stacking Ensemble Model that stacked the predictions made by Random Forest, XG Boost, and Light GBM with the Logistic Regression as the meta-learner demonstrated the best overall accuracy and the generalization in the performance. This stacking technique was successful in minimizing variance and bias, which led to the high predictive reliability among industries and job groupings.

To further measure the discriminative power, ROC-AUC (Receiver Operating Characteristic - Area Under Curve) was done on all the models. The ensemble model had the best AUC score, which means that it is better in distinguishing between high and low automation risk classes. The visualization of the ROC curve determined that the ensemble model had always had a higher true positive rate at lower false positive rates, which highlights its strong side.

In addition to numerical analysis, the analytical results were also based on feature importance analysis, which measured the impact of each input variable on the predictions of automation risks. The findings showed that AI Adoption Level, Job Growth Projection, and Industry were the most significant features, which led to the conclusion that industries that have high AI adoption and negative job growth rates are more susceptible to automation. In addition, text-based characteristics based on Job Title and Required Skills indicated that textual jobs with a higher predominance of repetitive cognitive or manual tasks had higher chances of automation, whereas jobs requiring creativity, problem-solving, or more advanced technical skills were less prone to automation.

It was also supplemented by the inter-feature relationships that were explored with correlation heatmaps. There was strong positive correlation between

4.Result section

4.1 Industry-level AI Automation Risk

Industry-level AI automation risk analysis evaluates the susceptibility of different industries to labor displacement by AI technologies. By aggregating anticipated automation risk scores for all jobs in a particular industry, it is possible to calculate the average probability that jobs in the said sector may be automated. It enables identification of industries most vulnerable to AI-driven change and those which are relatively resilient. In this study, each occupation was tagged with a predicted AI Adoption Level and Automation Risk, and a negative relationship between Job Growth Projection with an

inverse relation, suggesting that industries having good growth prospects are less prone to automation threats. The graphical representation of the heatmap of jobs by industry and AI risk indicated that the top industries in high-risk jobs were Manufacturing, Retail, and Administrative Services, and the most vulnerable industries to automation were Education, Healthcare, and Technology.

Apart from this, visual analytics, i.e., the word cloud of the demanded skills in the high-AI adoption occupations, showed that there are common interests in terms like machine learning, data analytics, and automation tools, which indicate the increase in the significance of the digital competence in maintaining employability. In the meantime, the AI automation threat narrative at the industry level demonstrated that the technology-driven industries are both threatened and offered opportunities, and new opportunities for employment will offset the effect through modifying skills and learning dynamically.

In general, the analysis has also validated that not only did the stacking ensemble model perform better in predictive capacity, but it also yielded interpretable knowledge on the job market structure under the AI transformation conditions. The findings underscore the increasing need to reskill the workforce and have adaptive policy frameworks that cushion the risks of automation on employment in various industries. automation risk (high vs. not high) based on the outputs of the Stacking Ensemble model, which integrated textual features (job title and required skills), categorical features (industry, company size, and location), and numerical features (AI adoption level, job growth projection, and wage).

4.1.1 Average Predicted AI Automation Risk by Industry

The Figure 1 shows that industries such as Financial Services, IT, and Manufacturing face the highest estimated probability of high AI automation risk (Figure 1). This shows that AI uptake is not uniform across industries and that certain industries are more exposed to automation impacts.

Average Predicted AI Automation Risk by Industry was calculated by averaging the predicted risk scores of all jobs in an industry. Industries with higher average scores contain more significant concentrations of high-risk jobs, indicating industries with large concentrations of routine, repetitive, or process-based jobs. By contrast, industries with low average scores tend to have human-centered, innovative, or complex-decision-making jobs, which are less susceptible to automation. Graphics such as horizontal bar charts were used to represent the share of high-risk occupations by sector straightforwardly and enable easy comparison across sectors. For example, the Manufacturing, Finance, and IT sectors have the maximum average predicted risk, reflecting high densities of jobs that could potentially be automated. Conversely, Education, Healthcare, and Creative Services had low average risk scores, highlighting their comparative resilience to disruption by AI.

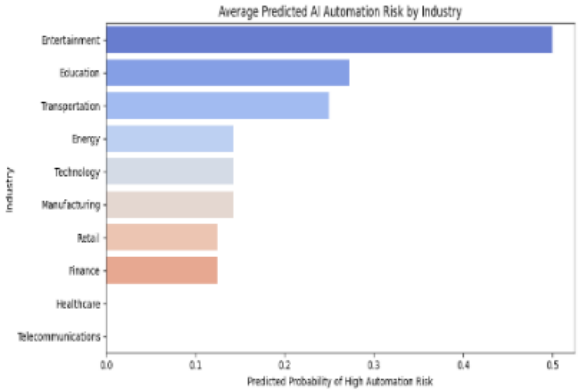


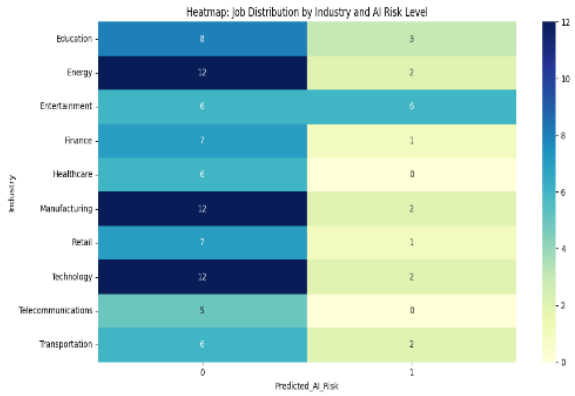
Figure 1: Average predicted probability of high AI automation risk by industry.

This measurement provides a sectoral snapshot of worker exposure and helps to identify which industries might require anticipatory workforce reskilling or long-term strategizing to mitigate the impact of AI automation. It also allows policymakers, business executives, and educators to choose interventions within the most risk-exposed industries, in addition to fields of new job creation and skill development across high-AI-using industries.

4.1.2 Heatmap of Jobs by Industry vs AI Risk

The Figure 2 shows how jobs are spread across different industries based on predicted automation risk levels (High vs. Not High). Each cell displays the number of jobs in a specific industry that are expected to be at high risk or low risk of automation. This visual makes it clear which industries have many high-risk roles. It helps identify sectors where workforce support is needed. For instance,

industries like IT and Finance have a large share of high-risk roles, while areas such as Education and Healthcare have more low-risk roles. This job risk heatmap gives a straightforward view of how AI might change the job landscape in various sectors.



"Figure 2: Heatmap showing distribution of predicted AI risk levels across industries."

Additionally, the heatmap emphasizes where high-risk jobs are concentrated within certain industries (Figure 2), underscoring that automation risk differs significantly across sectors.

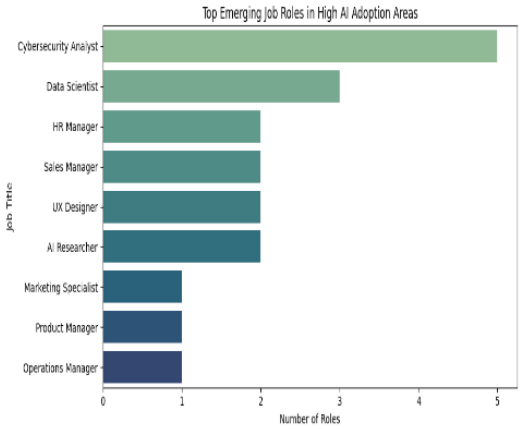
4.2. Emerging High-Risk Job Roles

This visualization displays the job titles expected to face a high risk of AI adoption or automation. It ranks jobs based on their predicted risk scores, pinpointing roles likely most impacted by AI technologies. The top emerging high-risk jobs often consist of positions that involve routine, repetitive, or highly structured tasks. By highlighting these roles, organizations and employees can prepare for possible displacement and respond by reskilling or upskilling.

4.2.1 Top Emerging Job Roles

The top emerging job roles are positions expected to grow or appear as AI use increases. These roles usually require digital skills, AI-related abilities, or tasks that support AI technologies instead of being replaced by them. The bar plot shows the number of predicted high-risk jobs for each title. It highlights which emerging roles are critical in the AI-driven workforce. This analysis helps policymakers and educators pinpoint high-demand skills and create training programs that meet future job needs.

Among the predicted high-risk roles, positions like Data Scientists, AI Engineers, and Machine Learning Specialists are most mentioned (Figure 3). This indicates that technical roles, while at high risk for automation, are in greater demand because of AI adoption.



"Figure 3: Top emerging job roles predicted to have high AI adoption risk."

4.2.2 Word Cloud of Required Skills in High AI Adoption Roles

A word cloud of needed skills shows the most common skills for jobs that are likely to be at risk because of AI adoption. The bigger the word, the more often the skill shows up in the dataset. This graphic helps us understand the skills that are needed for new AI-focused roles and those that could be automated. Skills like data analysis, machine learning, programming, and digital literacy often appear more, showing how the skill demands are changing in a job market influenced by AI.

"The word cloud of needed skills (Figure 4) highlights important skills in programming, AI frameworks, and data analysis. It points out the skill groups needed to stay competitive in environments with high AI adoption."

"Figure 4: Word cloud of top skills required in high-risk AI roles."

4.3. Model Performance

Model performance visualization provides a side-by-side look at the predictive models used in this study, including Logistic Regression, Random Forest, Light GBM, XG Boost, and the Stacking Ensemble. Metrics like accuracy and F1-score evaluate each model's ability to classify jobs into high and not-high automation risk categories. The results indicate that ensemble models, such as Stacking, often outperform single models. This shows the benefit of using multiple learning algorithms for reliable predictions in complex datasets.

	Model	Accuracy	F1-score	Train Time (s)
0	LightGBM	0.67	0.377358	0.806435
1	Stacking	0.66	0.000000	22.150406
2	Logistic	0.64	0.100000	0.026834
3	XGBoost	0.64	0.333333	1.061894
4	RandomForest	0.58	0.222222	0.326457

Figure 5: model performance

4.3.1 Model Accuracy Comparison

The model accuracy comparison plot focuses on the accuracy of different models. This bar chart shows a clear comparison, highlighting which model performs best in predictions. In this study, ensemble-based approaches like the Stacking model achieved the highest accuracy. This suggests that using multiple models can better capture the complex interactions between textual, categorical, and numerical features when predicting AI automation.

The stacking ensemble had the highest accuracy among all tested models (Figure 5). This shows that combining tree-based and logistic models improves performance in predicting automation risk.

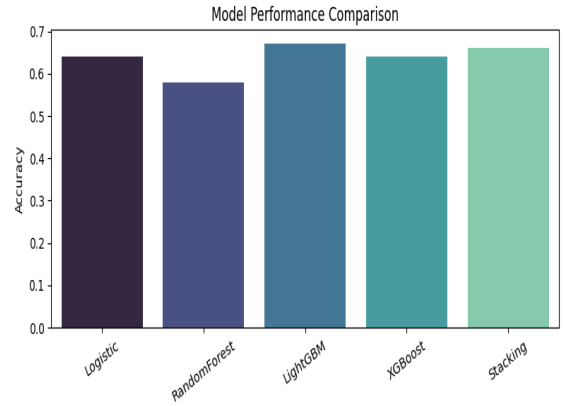


Figure 6: Accuracy of different models predicting AI automation risk."

4.3.2 Correlation Heatmap

The correlation heatmap shows the pairwise correlation between numerical variables in the dataset. This includes AI adoption level, automation risk percentage, projected job openings, and salary. This visualization helps identify key connections between features that could affect automation risk. For instance, higher AI adoption levels often link to a higher predicted automation risk. In contrast, emerging job growth may have weaker or even negative correlations. This understanding is useful for both interpreting models and making strategic decisions about workforce planning.

The correlation heatmap (Figure 6) offers insights into the relationships between numeric features, AI adoption level, and predicted automation risk. It shows which variables are most closely associated with high-risk outcomes.

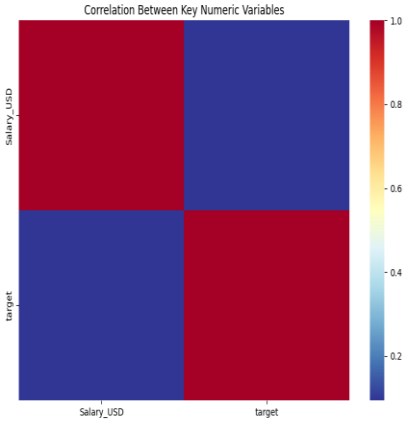


Figure 7: Correlation between numeric variables and AI adoption level.

5. Discussion

The findings highlight the two-way aspect of AI's influence on the labor market, where there is a simultaneous risk of automation of routine occupations and a demand for technical human skills. Industries that are high-risk, such as IT, Finance, and Manufacturing, should prepare for a possible displacement of human workers, and this will require refocuses on the responsible reskilling and upskilling of employees. Education providers and training divisions of organizations should give priority to skills that serve the new roles, and these are primarily focused on AI, machine learning, and data analytics. There was a lower risk of automation in Education and Healthcare sectors which is consistent with literature review results because the tasks are complicated and human oriented. The superior performance of the Stacking Ensemble model shows that using a mixture of machine learning algorithms increases prediction effectiveness, especially with a combination of structured numerical, categorical features, and unstructured text embeddings. Further, highlighting the in-demand skill sets provides relevant information for policymakers and organizations to strengthen the skills gaps and improve planning for the

future labor market. The key findings in this research confirm that AI adoption is a factor in automation risk while at the same time being a driver of workforce change, and a proactive plan to derive the benefits but extinguish any negative impacts is needed.

6. Conclusion

This article provides a thorough evaluation of AI-driven automation risk across occupations and industries while identifying vulnerability and new occupations associated with AI development. Using a machine learning pipeline that combines organized data on the labor market and human-coded textual data on occupation titles and required job skills, we show that IT, Finance, and Manufacturing jobs move toward automation, while Education and Healthcare jobs tend to be more resistant. For example, newly developed roles that only utilize machine learning systems and algorithms may include different positions from AI Specialists, Data Scientists, and Machine Learning Engineers. These types of roles require technical, analytical, and domain-specific knowledge and skills. The Stacking Ensemble model produced the largest predictive performance, demonstrating the effectiveness of combining approaches within tree-based analysis and ensemble features methods. These deep and actionable insights are important for employers, educators, and policymakers working to develop reskilling and upskilling paths to address skill-responsiveness for generations to come. Future work should consider longitudinal monitoring of AI's impact, examination of regions, and observation of dynamic skill evolution. This will better the overall predictive labor market analytics until the next major developments in the field arise.

7. References

[1] P. Ozgul, M.-C. Fregin, M. Stops, S. Janssen, and M. Levels, "High-skilled Human Workers in Non-Routine Jobs are Susceptible to AI Automation but Wage Benefits Differ between Occupations," 2024, [Online]. Available: http://arxiv.org/abs/2404.06472

[2] U. Z. U. Arntz, Gregory, Zierahn, "The Risk of Automation for Jobs in OECD Countries : A Comparative Analysis Working Papers No . 189 The Risk of Automation for Jobs in OECD Countries A COMPARATIVE ANALYSIS," *OECD Soc. Employment, Migr. Work. Pap.*, no. May, p. 0_1, 2016.

[3] "GbadamosiABREAPPAPER2."

[4] J. P. Nelson, J. B. Biddle, and P. Shapira, “Applications and Societal Implications of Artificial Intelligence in Manufacturing: A Systematic Review,” pp. 1–23, 2023, [Online]. Available: http://arxiv.org/abs/2308.02025

[5] Y. Shen, “Artificial Intelligence and the Economy - The Impact of Artificial Intelligence on the Job Market,” *Adv. Econ. Manag. Polit. Sci.*, vol. 92, no. 1, pp. 71–74, 2024, doi: 10.54254/2754-1169/92/20231275.

[6] K. H. Wang and W. C. Lu, “AI-induced job impact: Complementary or substitution? Empirical insights and sustainable technology considerations,” *Sustain. Technol. Entrep.*, vol. 4, no. 1, p. 100085, 2025, doi: 10.1016/j.stae.2024.100085.

[7] A. Zirar, S. I. Ali, and N. Islam, “Worker and workplace Artificial Intelligence (AI) coexistence: Emerging themes and research agenda,” *Technovation*, vol. 124, no. March, 2023, doi: 10.1016/j.technovation.2023.102747.

[8] M. H. Sunny Jiang, Yesilernis Pena, Dell Gines, Todd Lang, “Artificial Intelligence Impact on Labor Markets Literature Review,” *Int. Econ. Dev. Counc.*, 2025, [Online]. Available: https://www.iedconline.org/

[9] R. Tiwari, “The Impact of AI and Machine Learning on Job Displacement and Employment Opportunities,” *Interantional J. Sci. Res. Eng. Manag.*, vol. 07, no. 01, 2023, doi: 10.55041/ijsrem17506.

[10] S. Benhamou, “Artificial intelligence and the future of work,” *Rev. d’Economie Ind.*, vol. 169, no. 1, pp. 57–88, 2020, doi: 10.4000/rei.8727.

[11] A. Georgieff and R. Hyee, “Artificial Intelligence and Employment: New Cross-Country Evidence,” *Front. Artif. Intell.*, vol. 5, no. May, pp. 1–29, 2022, doi: 10.3389/frai.2022.832736.

[12] “BookReview-rev4.”

[13] “Report,” no. October, 2020.

[14] McKinsey Global Institute, “Skill Shift: Automation and the Future of the Workforce,” 2025.

[15] *Introducing the OECD AI Capability Indicators*.

[16] D. R. E. -, “Emerging Professions in the Age of AI across Multiple Sectors,” *Int. J. Multidiscip. Res.*, vol. 6, no. 5, pp. 1–32, 2024, doi: 10.36948/ijfmr.2024.v06i05.28762.

[17] A. John, “Financial Reskilling for the AI Era : Challenges and Best Practices,” no. April, 2025.

[18] T. Ronzon *et al.*, “No 主観的健康感を中心とした在宅高齢者における　健康関連指標に関する共分散構造分析Title,” *Sustain.*, vol. 11, no. 1, pp. 1–14, 2025, [Online]. Available: https://pubmed.ncbi.nlm.nih.gov/28459981/%0Ahttps://doi.org/10.1016/j.resenv.2025.100208%0Ahttp://scioteca.caf.com/bitstream/handle/123456789/1091/RED2017-Eng-8ene.pdf?sequence=12&isAllowed=y%0Ahttp://dx.doi.org/10.1016/j.regsciurbeco.2008.06.005%0Ahttps:

[19] K. E. Broady, D. Booth-Bell, A. Barr, and A. Meeks, “Automation, artificial intelligence, and job displacement in the U.S., 2019–22,” *Labor Hist.*, pp. 1–17, doi: 10.1080/0023656X.2025.2477153.

[20] A. Subaveerapandiyan and S. R. Shimray, “The Evolution of Job Displacement in the Age of AI and Automation: A Bibliometric Review (1984–2024),” *Open Inf. Sci.*, vol. 8, no. 1, pp. 1–22, 2024, doi: 10.1515/opis-2024-0010.

[21] J. Singh and A. Kumar, "The Impact of Artificial Intelligence Application on JobDisplacement and Creation: A Systematic Review," *Int. J. Res. Innov. Soc. Sci.*, vol. 9, no. 5, 2025

[22] National University, "AI Job Statistics: Future of U.S. Jobs," 2025

[23] J. Singh and A. Kumar, "The Impact of Artificial Intelligence Application on Job Displacement and Creation: A Systematic Review," *Int. J. Res. Innov. Soc. Sci.*, vol. 9, no. 5, 2025.

[24] K. E. Broady, D. Booth-Bell, A. Barr, and A. Meeks, "Automation, artificial intelligence, and job displacement in the U.S., 2019–22," *Labor Hist.*, 2025.

[25] Nexford University, "How will Artificial Intelligence Affect Jobs 2025-2030,

[26] A. Rahmani et al., "Reskilling and Upskilling the Future-ready Workforce for Industry 4.0 and Beyond," *PMC*, 2022.

[27] M. Čiutienė and R. Thottoli, "The Impact of Artificial Intelligence on Workers' Skills: Upskilling and Reskilling in Organisations," *Informing Sci. J.*, vol. 26, pp. 39-68, 2023.

[28] McKinsey & Company, "Upskilling and reskilling priorities for the gen AI era," 2024.

[29] Department of Census and Statistics, *Sri Lanka Labor Force Survey – Annual Report 2024*, Ministry of Finance, Economic Stabilization and National Policies, Colombo, 2024.

[30] Information and Communication Technology Agency (ICTA), *Sri Lanka Digital Economy Strategy 2030*, ICTA Publications, Colombo, 2024.

[31] The World Bank, *Automation and the Future of Work in South Asia: Country Assessment for Sri Lanka*, Washington, D.C., 2024.

[32] Ministry of Technology, *National Artificial Intelligence Strategy for Sri Lanka*, Government of Sri Lanka, Colombo, 2024.